

Staged Rule Acquisition in WishGLD: DISSOCIATION AND COMPRESSION ACROSS LEARNING

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1 BACKGROUND

Set shifting and rule acquisition:

In WCST-like tasks, the sorting rule changes without warning and must be re-inferred from feedback. Search cost, flexibility, and feedback-locked neural measures are all defined *relative to when the rule was acquired*. The acquisition marker is therefore not a detail: it is the anchor for everything measured downstream.

Core Question:

Is rule acquisition a discrete event or a gradual process?

The Divergence:

- (1) Learning as *a gradual updating* of dimension weights. [1], [2]
- (2) Learning as *all-or-none*, mastered in a single trial [3], [4], [5], [6]

Why this remain unsolved:

A single behavioral marker, the first correct response, is compatible with **both** accounts: each predicts that responding eventually becomes correct.

2 METHODS

Wisconsin-ish Global-Local Dissociation Task (WishGLD)

A WCST variant in which the sorting rule is defined at *two hierarchical levels*. The relevant rule is never instructed and must be inferred from feedback alone. Four rules are possible: object or colour at the *global* level, object or orientation at the *local* level.

Trial: See Fig.1 for details.

Blocks and sets:

The rule stays fixed within a block and changes without warning at each block boundary, shifting either *within* the same hierarchical level or *across* levels. 96 analysed blocks were divided into 8 sets of 12, providing an early-to-late learning axis.

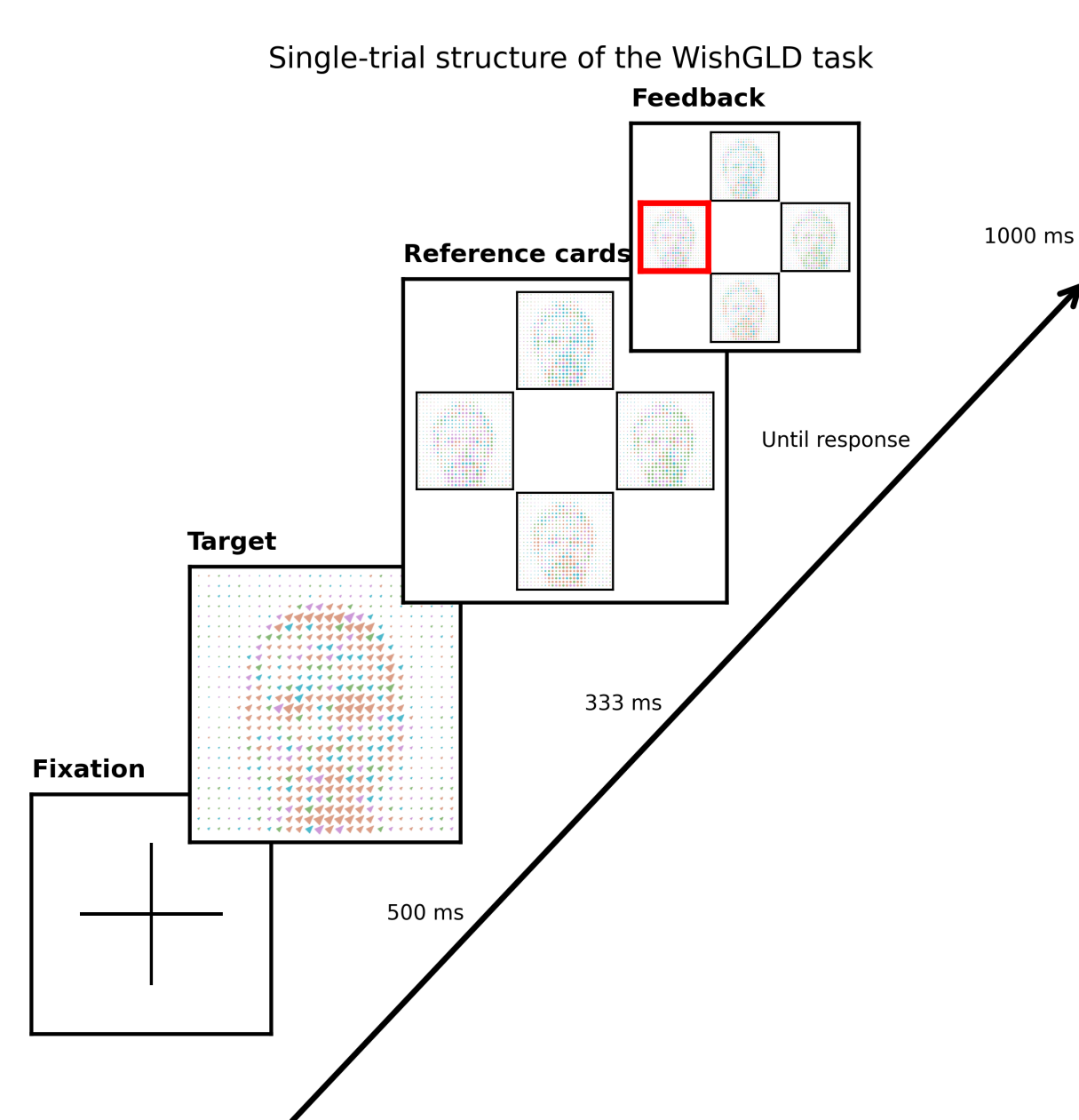


Fig.1:WishGLD paradigm. Fixation cross 500 ms → target card ~333 ms → four reference cards until response → feedback 1000 ms.

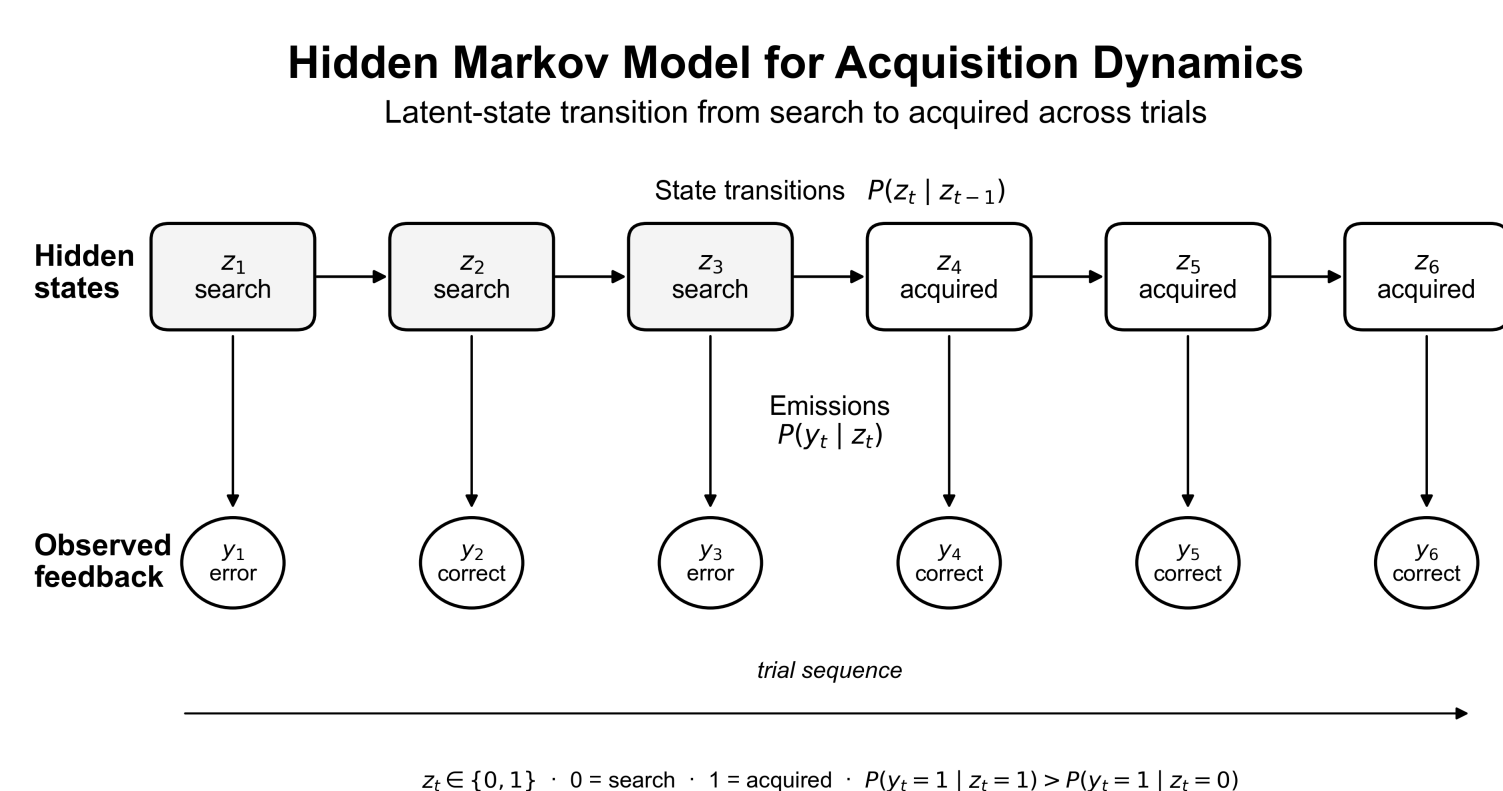


Fig.2:Unified two-state HMM used to derive behavioral transition markers from trial-wise correctness, plus lag/width geometry metrics. The model provides a structured behavioral timeline; mechanistic claims are validated with independent logical and neural markers.

Why HMM?

Search & Acquired - 2 states

Inner status (learning status) is hidden - latent state

Time sequenced, partially dependent - Markov

Learning is basically unidirectional and sticky - transition matrix

Transition (If any) - γ

Behavioral Markers

Logical Collapse Trial

Trial	Candidate rule space	Correct rule	Response	Feedback
1	A B C D	B	C	Error
2	A B C D	B	A	Error
3	A B C D	B	D	Error
4	A B C D	B	D	Error ← logical collapse
5	A B C D	B	C	Error
...				
11	A B C D	B	B	Correct ← first correct

Table 1: every different error feedback eliminates a rule in the rule space, unless there is only one candidate rule in the rule space, we mark this trial as collapse trial. Hypothetically, if subject learns fast enough, the collapse trial shall overlap with the first correct trial.

First Correct Trial and Acquisition Point

Here, we define the learning point as the first trial of the 2 consecutive trials with $\gamma > 0.8$.

Hypotheses

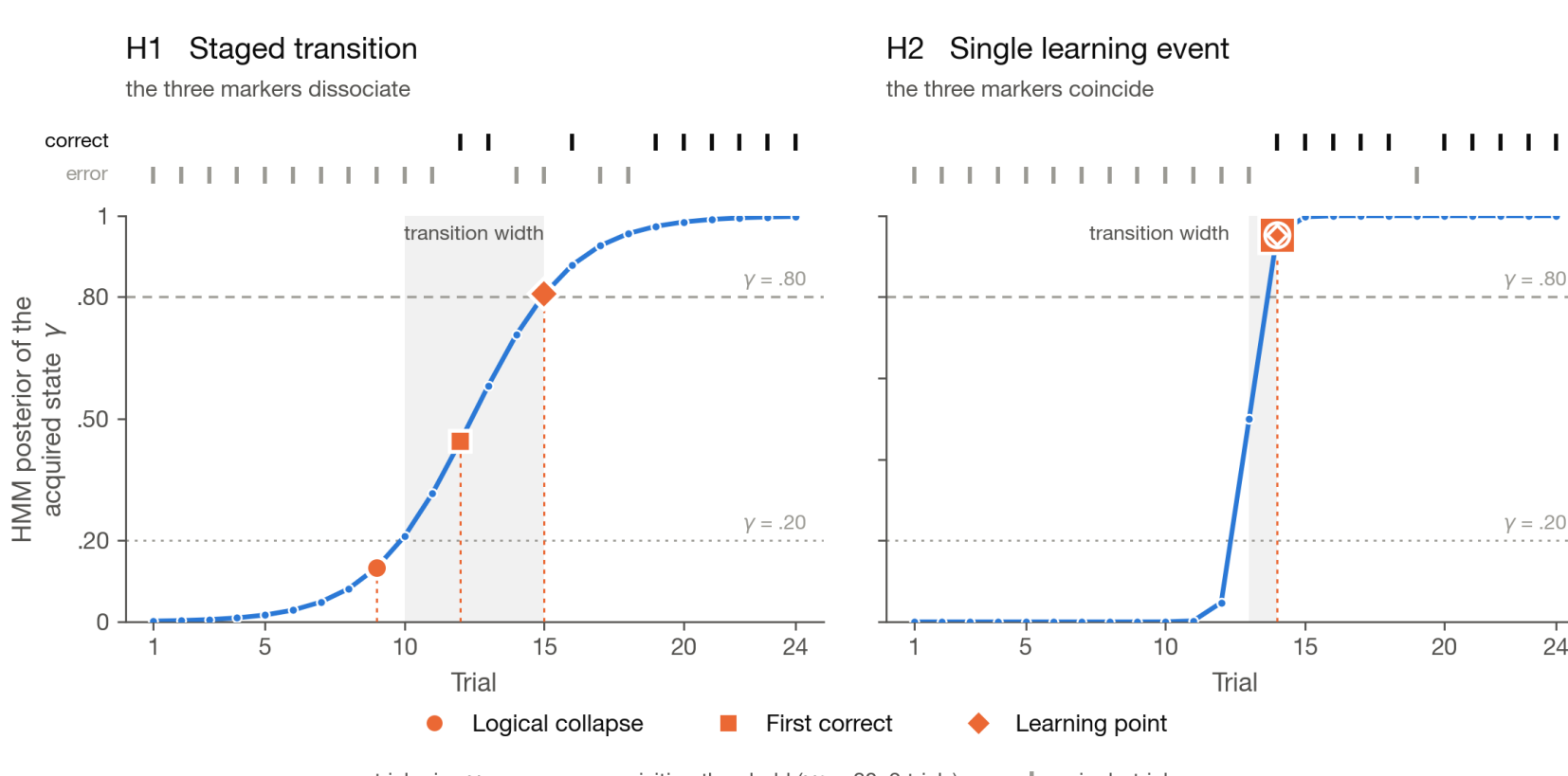


Fig.3: The learning curves and the relationship of the three markers for each of the two hypotheses. If the gradual updating is true, then the learning curve is smooth and the three markers should separate. For the all-or-none hypothesis, learning curve is supposed to be sharp, meanwhile the three markers are closer to each other. **Transition Width** refers to the number of trials subjects take from logical collapse point to the learning point.

3 RESULTS

Behavioral Results

Transition Width: the number of trials subjects take from logical collapse point to the learning point.

Acquisition lag: the number of trials subjects take from first correct response to the acquisition point.

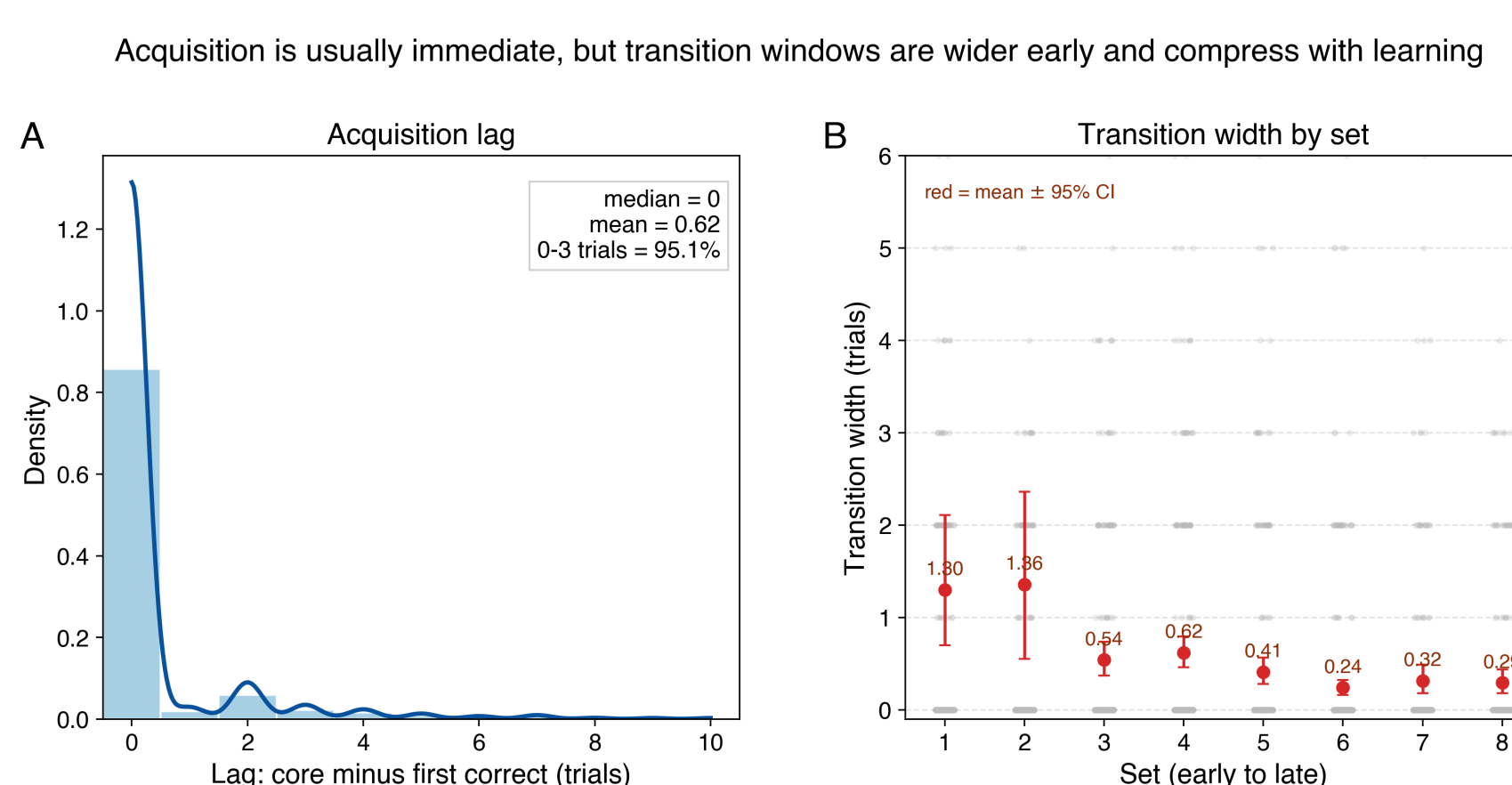


Fig.4: Acquisition is usually immediate (median lag = 0), yet the transition window was measurably wider in early sets (mean 1.3 trials) than late sets (mean 0.3), and compressed monotonically with experience.

EEG Results

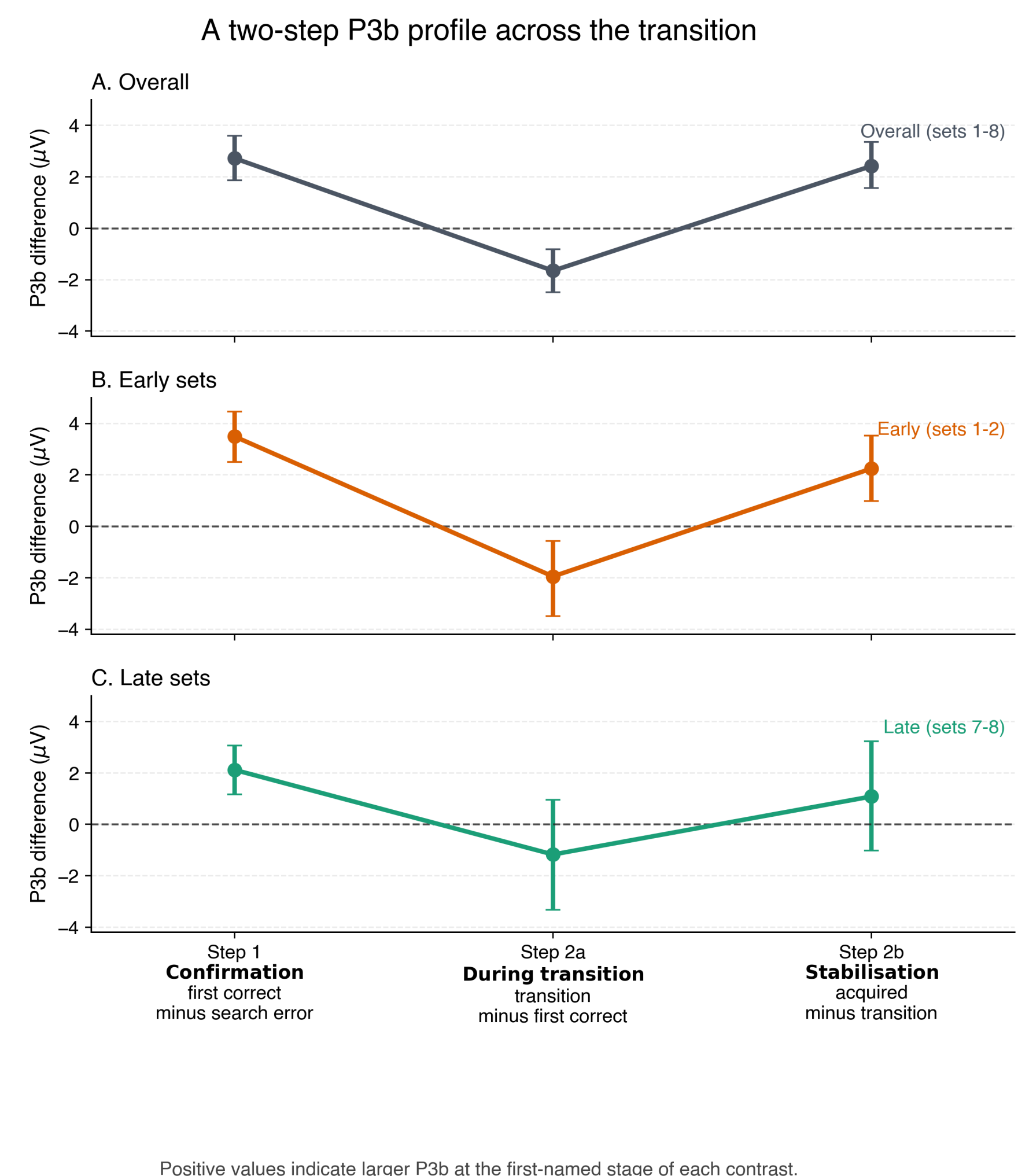


Fig.5: P3b rose at the first-correct boundary (Step 1, $\Delta = 2.72 \mu V$), fell during the transition (Step 2a, $\Delta = -1.65 \mu V$), and rose again at stabilisation (Step 2b, $\Delta = 2.42 \mu V$; all $\Pr(\text{direction}) \geq 0.999$). The profile was most pronounced in early sets and flattened in late sets, where only Step 1 remained reliable.

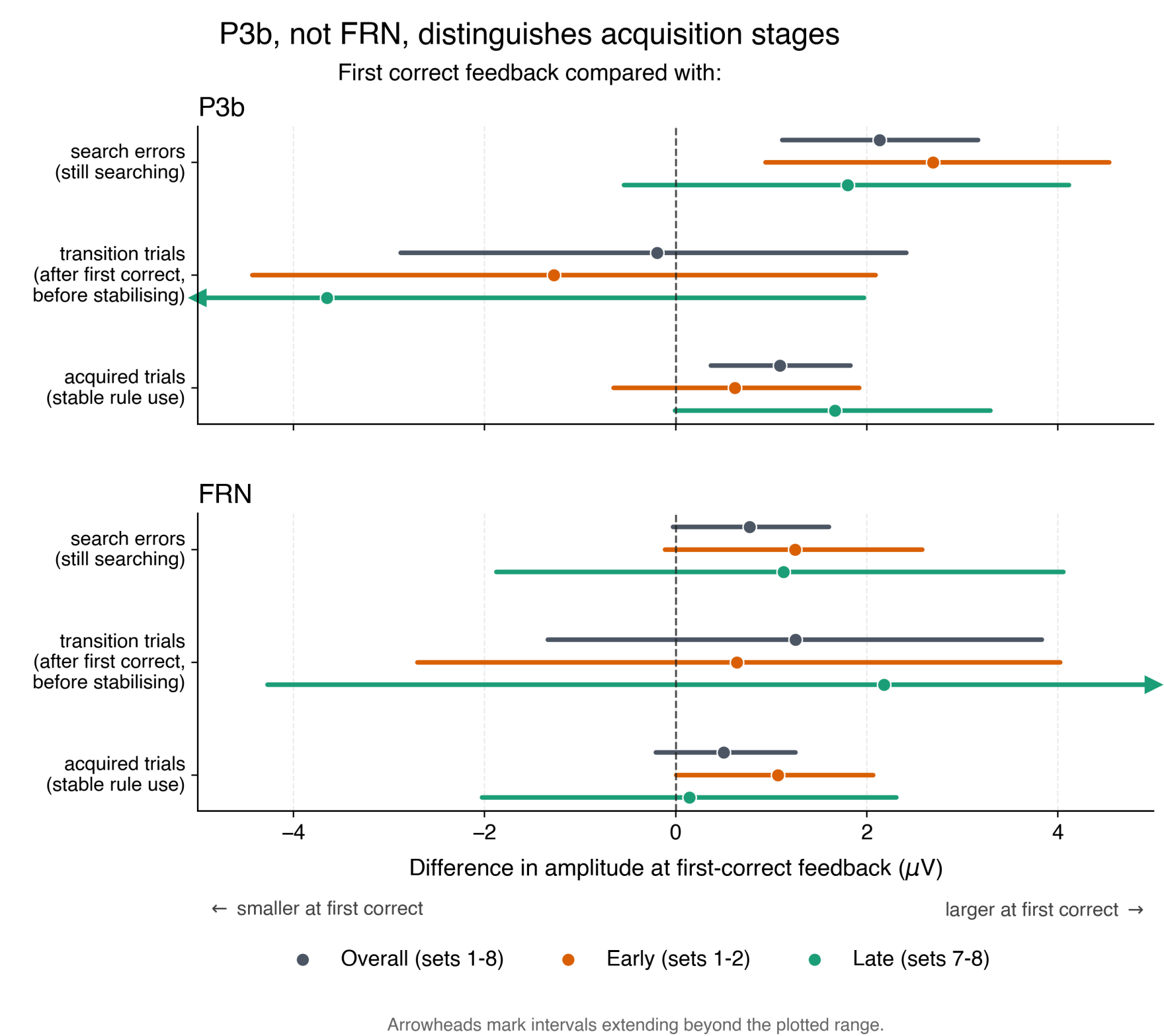


Fig.6: P3b was reliably enhanced at first-correct feedback relative to search errors ($\Delta = 2.14 \mu V$, 95% CrI [1.11, 3.17]) and relative to acquired feedback ($\Delta = 1.09 \mu V$, [0.37, 1.83]). FRN credible intervals crossed zero in all but one contrast. Points = posterior means, bars = 95% CrI; positive values indicate larger amplitude at first correct.

Conclusion

- **Neither account holds globally.** Acquisition was usually immediate (median lag = 0), yet early sets showed measurably wider transition windows that compressed with experience ($1.3 \rightarrow 0.3$ trials).
- **The markers dissociate.** Logical collapse never followed behavioral stabilization, and preceded it in 40% of blocks: having sufficient evidence is not the same as acting on it stably.
- **P3b and FRN play different roles.** P3b showed a staged up-down-up profile across the transition; FRN did not mark stages, but predicted how quickly stabilisation followed.
- **Implication.** A single behavioural marker conflates logically, neurally and behaviourally distinct operations. The discrete and gradual traditions may have sampled *different stages of the same process*.

References

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